$$\int_1^2 \frac{1}{\log^\alpha x}\,dx \sim \int_1^2 \frac{1}{(x-1)^\alpha}\,dx$$

$$c \cdot (x-1)^{1-\alpha} \qquad \alpha < 1$$

∞

$$\int_1 \frac{1}{x \log^\alpha x (e^x+1)^\beta}\,dx, \qquad \int_1^\infty \frac{1}{x (e^x+1)^\beta}\,dx$$

$$e^x + 1 \geq 1 + x$$